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Roll No

CE-701-GS

B.E. VII Semester

Examination, December 2020

Grading System (GS)

Design of Hydraulic Structure

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Explain factors governing design of barrages.
b) Design the practical profile of gravity dam of stone masonry, height of the dam is 30.0m, specific gravity of masonry = 2.4 and safe compressive stress for masonry of dam = 1200 kN/m^3 .
2. a) Describe various type of spillways with their suitability criteria.
b) Calculate discharge over a spillway crest length 100m, where in water depth changes from 1.0m to 4.0m in hydraulic jump formation in stilling basin.
3. What are various modes of failure of gravity dam? Also briefly explain their factor of safety.

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4.
 - a) Explain design considerations for fixing the section of dam.
 - b) Explain stability analysis method for gravity dam.

5.
 - a) Enlist various Hydro-power plants and describe any one in detail with sketch.
 - b) Write short notes on
 - i) Draft tube
 - ii) Surge tank

6.
 - a) Discuss the bilgh's creep theory with their limitation.
 - b) What are the main causes of failure of a weir in permeable foundation? What remedies would you suggest to prevent them?

7.
 - a) How are the load factor, capacity factor and utilization factor Interrelated what is signification of Diversity factor?
 - b) Discuss the relative merits of hydro electric plant as compared to other power resources.

8. Write short notes on any four of the following.
 - a) Rock fill Dams
 - b) Life of Reservoir
 - c) Weir and Barrage
 - d) Purpose of Galleries
 - e) Energy Dissipaters

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